KCE PATHWAYS · GRADES 4-5 · UNIT 3: BUILDING THINGS · WEEK 12 · SCIENCE

Energy and Electricity

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. What is energy? Name three things at home that use energy.

2. Draw a battery, a wire and a bulb connected so you think the bulb would light. Use arrows to show where you think electricity goes.

3. Predict: Which do you think lets electricity pass through: a metal spoon, a plastic spoon, or both? Why?

4. Where do you think the electricity in your home comes from before it reaches the wall?

5. What is one thing you wonder about electricity?

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Show What I Learned

Use evidence from today's lesson to explain your thinking.

1. Trace one energy chain from today (for example, a flashlight or a soccer kick). Name each form of energy, in order.

2. Explain what a circuit needs for a bulb to light, and what a switch does.

3. Use the circuit tester data: name two conductors and two insulators. Then explain the surprise about salt water or graphite.

4. Choose Lewis Latimer or Granville T. Woods. What problem did he solve, and how did his work help people?

★ **Challenge:** Design a 'safe cord' for a toy robot. Draw it and label which parts must be conductors and which must be insulators, and explain why for each part.

Looking Ahead: Next week we study forces and motion: pushes, pulls, friction, gravity and simple machines. Question to bring: Why does a ball rolling on grass stop sooner than one rolling on a smooth floor?